Bluestock Mutual Fund Analytics Platform

Final Capstone Report

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Organization

Bluestock Fintech Pvt. Ltd.

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Executive Summary

The Mutual Fund Analytics Platform was developed to provide a complete analytics solution for evaluating Indian mutual funds using publicly available data from AMFI India and mfapi.in. The project covers the entire analytics lifecycle, beginning with data ingestion and transformation, followed by database design, exploratory data analysis, risk and performance analytics, investor behavior analysis, dashboard development, and reporting.

The platform processes more than 46,000 NAV records, 32,000 investor transactions, benchmark index data, AUM statistics, SIP inflows, and portfolio holdings. Using Python, SQLite, Tableau/Power BI, and statistical analysis techniques, the project generates actionable insights into mutual fund performance, investor trends, and market behavior.

The final solution includes an automated ETL pipeline, analytical notebooks, advanced risk metrics, an interactive dashboard, and a recommendation engine.

1. Introduction

Mutual funds have become one of the most preferred investment vehicles in India. Increasing investor participation, growing SIP inflows, and rising industry AUM require data-driven methods to evaluate fund performance and investor behavior.

This project aims to build an end-to-end Mutual Fund Analytics Platform capable of:

- Processing large-scale mutual fund datasets
- Calculating risk and performance metrics
- Benchmarking funds against market indices
- Understanding investor demographics
- Creating interactive dashboards
- Generating actionable business insights

2. Problem Statement

Investors often face challenges while selecting mutual funds due to the availability of numerous schemes, varying risk levels, and complex performance metrics.

Key challenges include:

- Lack of centralized analytics
- Difficulty comparing funds
- Limited visibility into investor behavior
- Inadequate risk assessment
- Limited benchmark comparisons

The project addresses these challenges through a complete analytics platform.

3. Project Objectives

The following objectives were defined:

O1

Build a Python ETL pipeline.

O2

Design a normalized SQL star schema.

O3

Perform exploratory data analysis.

O4

Compute performance and risk metrics.

O5

Build a four-page interactive dashboard.

O6

Analyze investor demographics and transaction patterns.

O7

Benchmark fund returns against major indices.

O8

Document findings through reports and presentations.

4. Data Sources

The project utilizes ten datasets.

Fund Master

Contains scheme information including:

- Scheme code
- Fund house
- Expense ratio
- Benchmark
- Risk category

NAV History

Contains daily NAV data from January 2022 to May 2026.

Total Records:
46,000+

AUM by Fund House

Quarterly AUM information for top AMCs.

Monthly SIP Inflows

Tracks industry SIP growth.

Peak SIP inflow:

₹31,002 Crore (Dec 2025)

Category Inflows

Net inflows by mutual fund category.

Industry Folio Count

Growth from:

13.26 Crore → 26.12 Crore

Scheme Performance

Performance metrics for all schemes.

Investor Transactions

32,000+ transactions across India.

Portfolio Holdings

Fund holdings and sector allocations.

Benchmark Indices

Includes:

- Nifty 50
- Nifty 100
- Nifty Midcap 150
- BSE SmallCap

5. ETL Architecture

Extraction

Data was extracted from:

- CSV datasets
- AMFI sources
- mfapi.in aligned datasets

Transformation

Major transformations performed:

- Missing value handling
- Date standardization
- Data validation
- Duplicate removal
- Type conversion

Loading

Processed data was loaded into SQLite.

The database schema follows a star-schema design.

Tables include:

- dim_fund
- dim_investor
- fact_nav
- fact_aum
- fact_sip
- fact_transactions

6. Database Design

The database was designed to support analytics and dashboard reporting.

Benefits include:

- Reduced redundancy
- Faster analytical queries
- Easier dashboard integration
- Improved scalability

Primary relationships were established using:

- amfi_code
- investor_id
- date fields

7. Exploratory Data Analysis

Industry AUM Growth

Industry AUM increased significantly between 2022 and 2025.

Key finding:

Industry AUM exceeded ₹81 Lakh Crore by December 2025.

SIP Growth

Monthly SIP inflows displayed strong growth.

Key milestone:

₹31,002 Crore monthly SIP inflow.

Folio Growth

Total folios nearly doubled during the study period.

Growth:

13.26 Cr → 26.12 Cr

Category Analysis

Mid-cap and small-cap categories experienced stronger inflows than traditional large-cap funds.

8. Fund Performance Analysis

Several risk and return metrics were calculated.

CAGR

Used to evaluate annualized growth.

Sharpe Ratio

Measures risk-adjusted returns.

Higher Sharpe indicates better risk-adjusted performance.

Sortino Ratio

Measures downside risk-adjusted returns.

Alpha

Measures excess returns over benchmark expectations.

Beta

Measures market sensitivity.

Maximum Drawdown

Measures worst historical decline.

9. Risk Analysis

Value at Risk (VaR)

Historical VaR (95%) was computed for all schemes.

Purpose:

Estimate maximum expected loss under normal market conditions.

Conditional Value at Risk (CVaR)

Measures average loss beyond the VaR threshold.

Provides deeper downside-risk analysis.

Rolling Sharpe Ratio

A 90-day rolling Sharpe ratio was calculated to monitor changing risk-adjusted performance over time.

10. Investor Analytics

Demographic Analysis

Investor behavior was analyzed using:

- Age group
- State
- Income
- City tier

Key Findings

- Investors aged 25–35 dominate SIP participation.
- Tier-1 cities contribute the largest transaction volumes.
- Tier-2 participation is growing rapidly.

State Analysis

Major contributors include:

- Maharashtra
- Karnataka
- Tamil Nadu
- Gujarat

11. Advanced Analytics

Investor Cohort Analysis

Investors were grouped by first investment year.

Findings:

The 2024 cohort contributed the highest total investment amount.

SIP Continuity Analysis

Investors with six or more SIP transactions were evaluated.

Criteria:

Gap > 35 days

Status:

At-Risk Investor

Fund Recommendation Engine

Recommendations were generated based on:

- Risk appetite
- Sharpe Ratio
- Risk category

Outputs included Low, Moderate, and High-risk recommendations.

Sector Concentration Analysis

Herfindahl-Hirschman Index (HHI) was used.

High HHI indicates concentrated portfolios.

Low HHI indicates diversified portfolios.

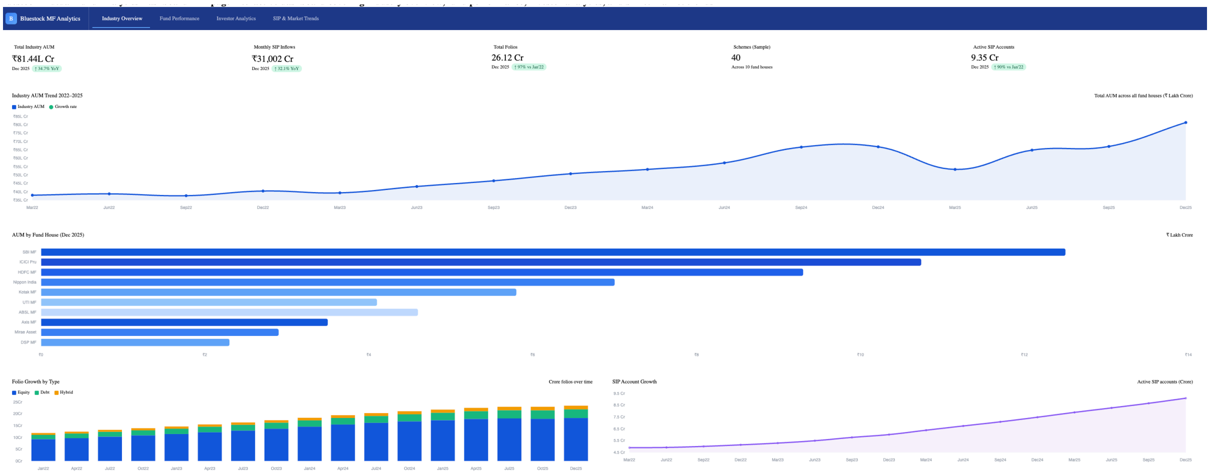
12. Dashboard Development

The dashboard consists of four pages.

Page 1: Industry Overview

Includes:

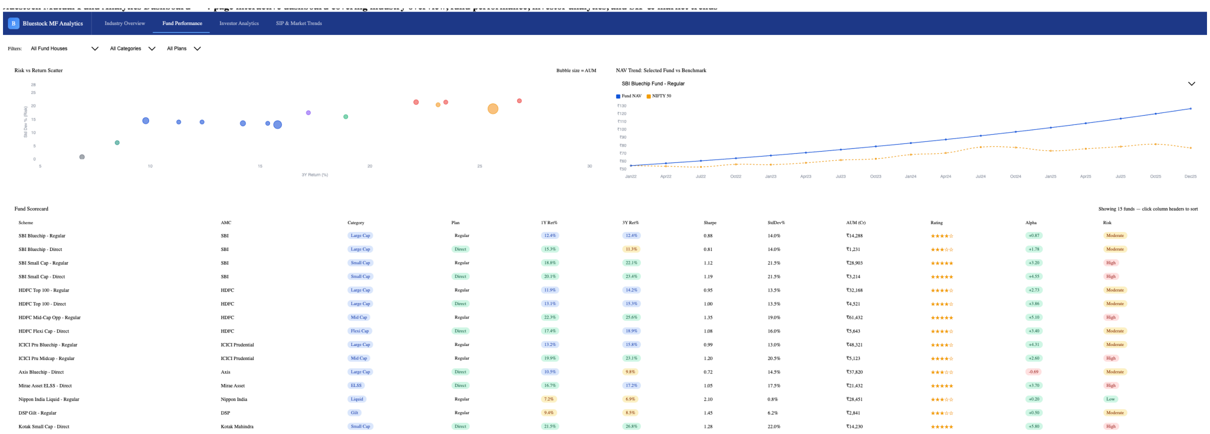
- Total AUM
- SIP Inflows
- Folio Count
- AUM by AMC



Page 2: Fund Performance

Includes:

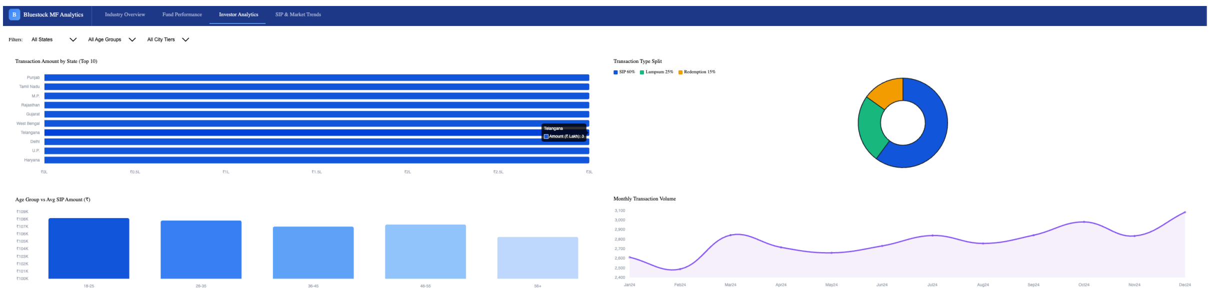
- Risk vs Return Scatter Plot
- Fund Scorecard
- NAV vs Benchmark



Page 3: Investor Analytics

Includes:

- State Analysis
- Transaction Types
- Age Group Analysis



Page 4: SIP & Market Trends

Includes:

- SIP vs Nifty
- Category Inflow Heatmap

- Market Trends

13. Key Findings

1. Industry AUM crossed ₹81 Lakh Crore.
2. SIP inflows reached ₹31,002 Crore.
3. Mid-cap funds generated higher alpha than large-cap funds.
4. High-risk funds achieved stronger Sharpe ratios.
5. The 2024 investor cohort invested the largest amount.
6. Several investors were flagged as SIP discontinuation risks.
7. Some equity funds showed high sector concentration.

14. Limitations

The project has the following limitations:

- Historical data constraints
- Simulated investor behavior assumptions
- Static recommendation methodology
- Limited live market integration

Future versions can improve these limitations.

15. Recommendations

1. Integrate live NAV APIs.
2. Implement machine learning recommendations.
3. Develop a Streamlit web application.
4. Add Monte Carlo simulations.
5. Deploy cloud-hosted dashboards.
6. Automate weekly reporting.

16. Future Enhancements

Potential enhancements include:

- Streamlit Dashboard
- Portfolio Optimization
- Efficient Frontier Analysis
- Automated Email Reporting
- Live Market Integration

17. Conclusion

The Bluestock Mutual Fund Analytics Platform successfully achieved all project objectives. The solution demonstrates the complete lifecycle of analytics engineering, including ETL development, database design, risk analytics, investor analytics, dashboard development, and reporting.

The platform provides a strong foundation for data-driven mutual fund analysis and showcases practical applications of data engineering, business intelligence, and financial analytics.